

# Asymmetric Capital Buffer Control

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## Abstract

A financial intermediary manages a capital buffer measured against a reference level, penalised asymmetrically for shortfall. This document states what the model establishes. Every numbered result carries a canonical verifier, recorded in Appendix A: a Lean proof where one exists, an exact symbolic check otherwise. The numerical findings of Section 7 are labelled as such and are findings at stated parameters, not theorems. Section 8 records what is not established.

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## Index of result codes

Every numbered result carries a three-letter code, used for reference in the accompanying summaries. Codes are stable; numbering may change.

| Code | Result   | Statement                                       |
|------|----------|-------------------------------------------------|
| ACI  | Prop. 1  | Accounting identity                             |
| PSA  | Prop. 2  | Persistence asymmetry                           |
| FPV  | Thm. 1   | Fourth-power variance ratio                     |
| ROB  | Prop. 3  | Robustness of the variance ratio                |
| EGE  | Prop. 4  | EGARCH-class equivalence                        |
| SJT  | Prop. 5  | Sojourn time                                    |
| NST  | Prop. 6  | Exact nesting of the risk-neutral benchmark     |
| RPC  | Prop. 7  | The running payoff is concave                   |
| CCP  | Prop. 8  | Consistency condition on parameters             |
| LQB  | Prop. 9  | Lower bound, the liquidation payoff             |
| TSO  | Prop. 10 | The two states are one state                    |
| CDA  | Cor. 1   | The degeneracy is a coordinate artefact         |
| SCG  | Prop. 11 | The surplus limit is cap geometry too           |
| REG  | Lem. 1   | Regularity in the parameters                    |
| ENV  | Lem. 2   | The envelope identity                           |
| BDR  | Lem. 3   | Differentiability of the boundaries             |
| TCS  | Prop. 12 | Threshold comparative statics                   |
| TCSN | Rem. 1   | What the proposition rests on                   |
| ENVK | Lem. 4   | The envelope identity for the fixed cost        |
| KCS  | Prop. 13 | Fixed-cost comparative statics                  |
| IDN  | Cor. 2   | Joint local identification                      |
| KCSN | Rem. 2   | Reading the identification result               |
| SOC  | Prop. 14 | Second-order conditions                         |
| SMF  | Prop. 15 | Smooth fit at the trigger                       |
| SMFN | Rem. 4   | Why the argument discriminates                  |
| SOCN | Rem. 3   | Strictness at the trigger                       |
| ITR  | Lem. 5   | Interior regularity                             |
| FIN  | Lem. 6   | No chattering                                   |
| VER  | Prop. 16 | Verification                                    |
| VERN | Rem. 5   | What verification needs                         |
| SLV  | Rem. 6   | The solver, and how it was validated            |
| CBI  | Rem. 7   | The cap binds throughout the inaction region    |
| CSL  | Rem. 8   | Comparative statics in the asymmetry parameter  |
| LNC  | Rem. 9   | Local non-concavity under a fixed issuance cost |
| KSW  | Rem. 10  | The fixed cost is the cause, a $K$ -sweep       |

**Terminology, binding throughout.** The parameter  $\lambda \geq 1$  is an *asymmetric risk-response parameter*. It is not called loss aversion. The distinction matters because  $\lambda = 1$  recovers a risk-neutral benchmark *exactly* (Proposition 6), so  $\lambda$  measures asymmetry in excess of what optimisation under regulatory constraint already produces — not in excess of zero. Loss-aversion language would additionally require elicited-choice evidence of the kind observational data cannot supply.

**Two parameters share the symbol  $\lambda$ .** The discrete-time variance ratio of Section 1 is written  $\lambda_V$ ; the shortfall penalty of Section 2 is written  $\lambda_S$ . They are distinct objects on the same state (Proposition 10), and no relation between them is asserted here; see Section 8.

# 1 The discrete-time model

## 1.1 Setup

Let  $E_t$  denote book equity and  $Q_t$  required capital, with  $Q_t$  *exogenous* and growing deterministically,  $Q_t = (1 + g)Q_{t-1}$ ,  $g > -1$ . The state is the log capital-buffer ratio

$$z_t = \log\left(\frac{E_t}{Q_t}\right), \quad (1)$$

so  $z_t = 0$  exactly when equity sits at its requirement;  $z_t > 0$  is surplus and  $z_t < 0$  is deficit. Equity accumulates by retention,  $E_t = E_{t-1}(1 + r_E b)$  where  $r_E$  is the return on equity and  $b \in (0, 1]$  the retention ratio.

Shocks enter with a state-dependent conditional variance,

$$z_t = \phi z_{t-1} + \varepsilon_t, \quad \text{Var}(\varepsilon_t \mid z_{t-1}) = V(z_{t-1}) = V_0 \exp(-2(\log \lambda_V) \tanh(z_{t-1}/\theta)), \quad (2)$$

with  $\theta > 0$  a saturation scale. The mechanism is reduced-form: trading limits tighten as the buffer erodes, so the same shock has a larger conditional variance in deficit than in surplus. Nothing in (2) is derived from an optimisation; that is the role of Section 2 onward.

## 1.2 Results

**Proposition 1** (ACI: Accounting identity). *Balanced growth of equity and requirement forces  $r_E = g/b$ .*

*Proof.* From  $E_t = E_{t-1}(1 + r_E b)$  and  $Q_t = Q_{t-1}(1 + g)$ ,

$$\frac{E_t}{Q_t} = \frac{E_{t-1}}{Q_{t-1}} \cdot \frac{1 + r_E b}{1 + g}. \quad (3)$$

Balanced growth means  $E_t/Q_t = E_{t-1}/Q_{t-1}$  for every  $t$  along the deterministic path, i.e. the factor multiplying  $E_{t-1}/Q_{t-1}$  in (3) equals 1:

$$\frac{1 + r_E b}{1 + g} = 1 \iff 1 + r_E b = 1 + g \iff r_E b = g \iff r_E = \frac{g}{b}. \quad (4)$$

Each step of (4) is reversible, so this is not merely necessary but exactly characterises balanced growth: given  $b \in (0, 1]$ ,  $r_E = g/b$  is well-defined for every  $g > -1$  and is the unique return on equity for which (3) holds with equality at every  $t$ .  $\square$

**Proposition 2** (PSA: Persistence asymmetry). *The deficit-side persistence is  $\phi^- = 1 - g(1 - b)/b$ , and  $\phi^- \rightarrow 1$  exactly as  $g \rightarrow 0$ . The surplus-side persistence  $\phi^+$  differs, giving an asymmetry  $\phi^- - \phi^+$  which vanishes with  $g$ .*

**Theorem 1** (FPV: Fourth-power variance ratio). *Under (2), the ratio of conditional variances deep in deficit to deep in surplus is*

$$\frac{\lim_{z \rightarrow -\infty} V(z)}{\lim_{z \rightarrow +\infty} V(z)} = \frac{V_0 \lambda_V^2}{V_0 \lambda_V^{-2}} = \lambda_V^4. \quad (5)$$

**Proposition 3** (ROB: Robustness of (5)). *The ratio depends only on the saturation limits of  $\tanh$ , not on  $\theta$ , and is therefore invariant to the shape of the transition between regimes.*

**Proposition 4** (EGE: EGARCH-class equivalence). *The recursion (2) is observationally equivalent to an EGARCH-class specification (exponential link, bounded, asymmetric) in  $\log V$ , conditioned on the state level  $z_{t-1}$  rather than on a standardised residual as in Nelson’s (1991) canonical EGARCH. Given  $\theta$ ,  $\lambda_V$  is estimable from returns data without observing  $Q_t$ .*

**Proposition 5** (SJT: Sojourn time). *For a buffer with drift  $\mu > 0$  started a distance  $d$  below the requirement, the expected first-passage time to  $z = 0$  is  $\mathbb{E}[T] = d/\mu$ , with the comparative statics in  $d$  and  $\mu$  immediate.*

Theorem 1 is the model’s one directly estimable implication. It is Lean-verified and depends on nothing in Sections 2–7.

## 2 The continuous-time formulation

### 2.1 State, controls, objective

Let  $X_t$  be the capital buffer and  $R$  a reference level. Three controls of three distinct mathematical types act simultaneously:

- $\pi_t$ , a *classical* control of risky exposure, constrained by a state-dependent regulatory cap  $\pi \leq \bar{\pi}(x)$ ;
- $D_t$ , a *singular* control of dividends, acting at an upper barrier;
- $I_t$ , an *impulse* control of recapitalisation, with cost  $K + (1 + \kappa)\xi$  for an injection of size  $\xi$  — a fixed cost *and* a proportional cost.

Shortfall below the reference is penalised at rate

$$\Lambda(x) = (\lambda_S - 1)(R - x)^+, \quad \lambda_S \geq 1. \quad (6)$$

The state is the assets-to-liabilities ratio  $x = A/L \in (1, \infty)$ , and  $V$  is the value *per unit of liabilities*: if  $J(A, L)$  denotes the unreduced value, then  $J(A, L) = L V(x)$ . This normalisation is what makes the problem one-dimensional, and it has two linked consequences for the equation below. Since  $\partial J / \partial L = V - xV'$ , the liability drift contributes  $\mu_L L(V - xV')$  to the unreduced generator; dividing by  $L$ , the term  $-\mu_L xV'$  joins the drift — which is why (9) carries  $-\mu_L x$  — and the discount term becomes

$$-\rho V + \mu_L V = -\rho_L V, \quad \rho_L := \rho - \mu_L, \quad (7)$$

so the *liability-net* rate  $\rho_L$ , not the gross rate  $\rho$ , discounts the reduced problem. Both halves of the reduction are forced together: an equation carrying  $-\mu_L x$  in its drift must carry  $-\rho_L V$  in its discount term.

Writing  $\mathcal{L}^\pi$  for the controlled generator of  $x$ , the value function satisfies the quasi-variational inequality

$$\max \left\{ \sup_{\pi \leq \bar{\pi}(x)} \mathcal{L}^\pi V(x) - \rho_L V(x) - \Lambda(x), \quad 1 - V'(x), \quad \mathcal{M}V(x) - V(x) \right\} = 0, \quad (8)$$

with  $\mathcal{M}V(x) = \sup_{\xi > 0} \{V(x + \xi) - K - (1 + \kappa)\xi\}$ . The drift and diffusion are

$$\mu(x, \pi) = x((1 - \pi)r + \pi\mu_s - \mu_L) + \gamma, \quad (9)$$

$$\sigma^2(x, \pi) = \pi^2 \sigma^2 x^2 + 2\pi c \sigma \sigma_L x(1 - x) + \sigma_L^2 (1 - x)^2, \quad (10)$$

where  $c$  is the correlation between the risky return and the liability shock, and  $\mu_L = \gamma + r_L$  combines liability growth and the rate paid on liabilities.

The supremum in (8) is taken over the constrained set  $\pi \leq \bar{\pi}(x)$ , so the formulation already accommodates both a binding and a slack cap: where the unconstrained maximiser lies below  $\bar{\pi}(x)$  the supremum takes it, and where it does not the supremum takes  $\bar{\pi}(x)$ . No separate boundary condition is attached to the switch between the two.

## 2.2 Two structural properties

**Proposition 6** (NST: Exact nesting of the risk-neutral benchmark).  $\Lambda(x) \equiv 0$  *identically at*  $\lambda_S = 1$ . *The problem therefore nests the risk-neutral benchmark exactly, rather than approaching it in a limit.*

**Proposition 7** (RPC: The running payoff is concave).  $-\Lambda$  *is concave on*  $\mathbb{R}$  *for every*  $\lambda_S \geq 1$ .

Proposition 7 marks the departure from reference-dependent models built on S-shaped utility, which are convex below the reference and therefore require a concavification argument before any threshold structure can be recovered: (6) is kinked-linear rather than S-shaped, so the running payoff stays concave. The claim should be stated at exactly that strength and no further. It is a statement about  $\Lambda$ , and it does *not* deliver a concave  $V$ : Remark 9 finds  $V$  locally convex above the recapitalisation boundary, caused by the fixed issuance cost  $K$  and present already at  $\lambda_S = 1$ . The difference from the S-shaped literature is that the non-concavity here is localised and cost-driven rather than global and preference-driven.

## 2.3 Coefficients where the cap binds

Where the regulatory cap binds,  $\pi = \bar{\pi}(x)$  and the supremum in (8) is attained at the constraint, so the generator's coefficients are

$$A(x) = \frac{1}{2} \sigma^2(x, \bar{\pi}(x)), \quad B(x) = \bar{\pi}(x) x (\mu_s - r) + (r - \mu_L)x + \gamma, \quad (11)$$

with  $\sigma^2$  as in (10). Write  $u(x) = \bar{\pi}(x)x$  for the capped exposure,  $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ , and let  $\bar{x}$  denote the level at which the two branches cross. On the solvency regime  $u(x) = (x-1)/a_1$  and (10) gives  $\sigma^2(x, \bar{\pi}(x)) = \kappa_1(c)^2(x-1)^2$  with

$$\kappa_1(c)^2 = \frac{\sigma^2}{a_1^2} - \frac{2c\sigma\sigma_L}{a_1} + \sigma_L^2. \quad (12)$$

**Proposition 8** (CCP: Consistency condition on parameters).  $u(1) = 0$ , so  $B(1) = (r - \mu_L) + \gamma = r - r_L$  under  $\mu_L = \gamma + r_L$ . A positive drift at the distress boundary therefore requires  $r > r_L$ .

The parameters used throughout satisfy this:  $r = 0.025$ ,  $\mu_L = 0.03$ ,  $\gamma = 0.02$ , giving  $r_L = 0.01$  and  $r - r_L = +0.015 > 0$ .

**Proposition 9** (LQB: Lower bound, the liquidation payoff).  $V(x) \geq x - 1$  for every  $x \geq 1$ ; in particular  $V(1) \geq 0$ .

*Proof.* The strategy that pays an immediate lump-sum dividend of  $x - 1$ , driving the state to the bankruptcy point, is admissible and collects exactly  $x - 1$ . Because the action is instantaneous, the running penalty  $\int e^{-\rho_L t} \Lambda(X_t) dt$  accrues nothing:  $\Lambda$  enters the objective only, never the admissible set, so the bound is unaffected by  $\lambda_S$ . Since  $V$  is a supremum over admissible strategies,  $V(x) \geq x - 1$ .  $\square$

### 3 The state variable, and the boundary at $x = 1$

Section 1 uses  $z = \log(E/K)$ : one special point ( $z = 0$ ), no finite lower boundary, equity exhaustion at  $z = -\infty$ . Section 2 uses a level ratio of assets to liabilities, where  $x = 1$  means equity exactly zero and the diffusion (10) degenerates. These look like different variables, with two special points rather than one and a finite degenerate boundary rather than none. They are not.

**Proposition 10** (TSO: The two states are one state). *Let the capital requirement be a ratio requirement,  $Q = qL$ . Then  $E/L = x - 1$  and  $E/Q = (x - 1)/q$ , so*

$$z = \log \frac{x-1}{q}, \quad x = 1 + q e^z, \quad (13)$$

a diffeomorphism of  $(1, \infty)$  onto  $\mathbb{R}$ . Under it:

- (i) the special points coincide —  $x = 1 \leftrightarrow z = -\infty$ ,  $x = 1 + q \leftrightarrow z = 0$ ,  $x \rightarrow \infty \leftrightarrow z \rightarrow \infty$ ;
- (ii) on the solvency regime the volatility of  $z$  is the constant  $\kappa_1(c)$  of (12), for every correlation  $c$ ;
- (iii) the drift of  $z$  is  $B(x)/(x-1) - \kappa_1^2/2$ , and  $B(1) = r - r_L$  by Proposition 8, so it tends to  $+\infty$  as  $z \rightarrow -\infty$  precisely when  $r > r_L$ .

**Corollary 1** (CDA: The degeneracy is a coordinate artefact, and the boundary is unreachable). *By (ii) the vanishing of the diffusion at  $x = 1$  is exactly cancelled by the blow-up of  $dz/dx$ : in  $z$  the diffusion is non-degenerate everywhere, so the degeneracy is a property of the imported coordinate, not of the model. By (iii), in  $z$ -coordinates the process has constant volatility and a restoring drift growing without bound as  $z \rightarrow -\infty$  whenever  $r > r_L$ ; equity is therefore never exhausted in finite time, and  $x = 1$  is unreachable. Unreachability is coordinate-invariant —  $x = 1$  is unreachable if and only if  $z = -\infty$  is — while its presentation as a degenerate-diffusion problem is not.*

Two limitations, stated rather than buried. The map requires reading Section 1's exogenous  $K$  as a *ratio* requirement set by regulation rather than a deterministic constant; with  $K$  constant and  $L$  stochastic no deterministic map exists. And (ii) holds on the solvency regime only — above  $\bar{x}$  the cap switches to  $(x - a_2)/a_3$  and the  $z$ -volatility is not constant. That range covers where the boundary question lives, since the recapitalisation boundaries of Section 7 all lie below  $\bar{x}$ , but it is a restriction, not a global result.

One consequence is worth recording because it is what makes  $\lambda_V$  and  $\lambda_S$  comparable at all: at the numerical baseline  $R = 1.15$  the requirement level is  $q = R - 1 = 0.15$ , and  $z = 0$  maps to  $x = 1.15 = R$  exactly. The two sections' reference points are the same point.

**Proposition 11** (SCG: The surplus limit is cap geometry too). *On the liquidity regime ( $x > \bar{x}$ ,  $u(x) = (x - a_2)/a_3$ ), the  $z$ -volatility tends, as  $x \rightarrow \infty$ , to the constant*

$$\kappa_3(c)^2 = \frac{\sigma^2}{a_3^2} - \frac{2c\sigma\sigma_L}{a_3} + \sigma_L^2, \quad (14)$$

the same functional form as  $\kappa_1(c)^2$  of (12) with  $a_1$  replaced by  $a_3$ . Consequently

$$\lambda_V^4 = \frac{\kappa_1(c)^2}{\kappa_3(c)^2}, \quad (15)$$

a function of  $(\sigma, \sigma_L, a_1, a_3, c)$  only.

*Proof.*  $\bar{\pi}(x) \rightarrow 1/a_3$  as  $x \rightarrow \infty$ ; substituting into (10) and taking the limit gives (14) directly, by the same computation as Proposition 10(ii) with  $a_1 \rightarrow a_3$ . Equation (15) is Theorem 1's definition evaluated at the two saturation limits.  $\square$

The content of Proposition 11 is what (15) does *not* contain: no  $\lambda_S$ ,  $K$ ,  $\kappa$ ,  $R$ ,  $\rho$ ,  $r$ ,  $\mu_s$ ,  $\mu_L$  or  $\gamma$ . Evaluated at the saturation limits,  $\lambda_V$  is a function of cap geometry and correlation alone. At the numerical baseline  $\kappa_1 = 1.7720$  and  $\kappa_3 = 0.2623$ , giving  $\lambda_V^4 = 45.633$  and  $\lambda_V = 2.599$  — a constant, carrying no information about  $\lambda_S$ . Two qualifications bound the reach of this. The limits are approached only as  $x \rightarrow 1^+$  and  $x \rightarrow \infty$ , and the process's actual range under the solved policy sits far from both: on a  $y_{\max} = 2.5$  grid the surplus-side approach to  $\kappa_3$  is still an order of magnitude short at the upper edge. And (15) is the *definitional* evaluation of  $\lambda_V$ ; it does not preclude a relation between an *estimated*  $\lambda_V$  and  $\lambda_S$  (Section 8).

## 4 Threshold comparative statics

Both boundaries respond to  $\lambda_S$ , and both responses reduce to properties of a single object: the expected discounted shortfall incurred under the optimal policy,

$$W(x) := \mathbb{E}_x \left[ \int_0^\infty e^{-\rho L t} (R - X_t)^+ dt \right], \quad (16)$$

the expectation taken under the policy optimal at the prevailing  $\lambda_S$ . Because  $\lambda_S$  enters the objective only through  $\Lambda(x) = (\lambda_S - 1)(R - x)^+$ , and enters it linearly,  $W$  is exactly the sensitivity of value to the preference parameter.

**Lemma 1** (REG: Regularity in the parameters). *Let  $\theta$  denote either  $\lambda_S$  or  $K$ , the other being held fixed. Then for every  $x$ :*

- (i)  $\theta \mapsto V(x; \theta)$  is convex and non-increasing;
- (ii) it is therefore locally Lipschitz, admits one-sided derivatives at every  $\theta$ , and is differentiable at all but countably many  $\theta$ ;
- (iii) at every  $\theta$  the one-sided derivatives are  $\partial_\theta^+ V = -\min_{a \in \mathcal{A}^*(\theta)} T(x, a)$  and  $\partial_\theta^- V = -\max_{a \in \mathcal{A}^*(\theta)} T(x, a)$ , where  $\mathcal{A}^*(\theta)$  is the set of optimal policies and  $T = S$  for  $\theta = \lambda_S$ ,  $T = N$  for  $\theta = K$ ; in particular  $V$  is differentiable at  $\theta$  if and only if every optimal policy yields the same  $T$ ;
- (iv)  $W(x; \cdot)$  is non-increasing in  $\lambda_S$  and  $N(x; \cdot)$  is non-increasing in  $K$ .

*Proof.* For fixed  $a$ ,  $\theta \mapsto J(x, a; \theta)$  is affine with slope  $-T(x, a) \leq 0$ : for  $\lambda_S$  this is the decomposition  $J = A - (\lambda_S - 1)S$  used below, and for  $K$  it is  $J = B - KN$ . Both are exact because the parameter multiplies a single term of the objective and neither restricts the admissible set.  $V(x; \cdot)$  is the pointwise supremum of this family, and a supremum of affine functions is convex; the slopes are non-positive, so the supremum is non-increasing. This gives (i), and (ii) is standard for a finite convex function on an interval. Part (iii) is the subdifferential formula for a supremum of affine functions,  $\partial_\theta V = -\text{co}\{T(x, a) : a \in \mathcal{A}^*(\theta)\}$ , whose endpoints are the stated one-sided derivatives; differentiability is the statement that this interval is a single point. Part (iv) is (i) restated: convexity means  $\partial_\theta V$  is non-decreasing, and  $\partial_\theta V = -T$  at points of differentiability, so  $T$  is non-increasing.  $\square$

Part (iv) is a testable by-product rather than a technical step, and the solve bears it out: across the  $\lambda_S$  sweep the second divided differences of  $V$  in  $\lambda_S$  are strictly positive at every grid point, and  $W$  is strictly decreasing in  $\lambda_S$  at every grid point. Neither was imposed on the solver.

**Lemma 2** (ENV: The envelope identity). *At any  $\lambda_S$  at which  $V(x; \cdot)$  is differentiable — by Lemma 1(ii), all but countably many — and with the optimal policy measurable in  $(x, \lambda_S)$ ,*

$$\frac{\partial V}{\partial \lambda_S}(x) = -W(x), \quad (17)$$

and  $W$  satisfies, on the inaction region,

$$\frac{1}{2}\sigma^2(x, u(x))W''(x) + \mu(x, u(x))W'(x) - \rho_L W(x) + (R - x)^+ = 0. \quad (18)$$

*Proof.* For an admissible policy  $a$  write the payoff as  $J(x, a; \lambda_S) = A(x, a) - (\lambda_S - 1)S(x, a)$ , where  $S(x, a) = \mathbb{E}_x[\int_0^\infty e^{-\rho_L t}(R - X_t)^+ dt]$  is the discounted shortfall under  $a$  and  $A$  collects the terms not involving  $\lambda_S$  — dividends received and impulse costs paid. The decomposition is exact because  $\Lambda$  is linear in  $\lambda_S$  and enters the objective only, never the admissible set (as in the proof of Proposition 9). Thus  $\lambda_S \mapsto J(x, a; \lambda_S)$  is affine with slope  $-S(x, a)$  for every fixed  $a$ , and  $V$  is their pointwise supremum. Under the stated hypotheses Danskin's theorem applies and the derivative of the supremum is the partial derivative evaluated at the maximiser, giving (17) with  $W(x) = S(x, a^*)$ .

For (18), differentiate the interior equation  $\sup_\pi \mathcal{L}^\pi V - \rho_L V - \Lambda = 0$  with respect to  $\lambda_S$ . The supremum is attained at  $u(x)$ , so by the same envelope argument the derivative of the supremum term is  $\mathcal{L}^{u(x)}(\partial V / \partial \lambda_S)$ ; and  $\partial \Lambda / \partial \lambda_S = (R - x)^+$ . Substituting  $\partial V / \partial \lambda_S = -W$  and negating gives (18).  $\square$

The two smooth-fit conditions then convert (18) into boundary data for  $W$ . At the payout barrier,  $V'(y^*(\lambda_S); \lambda_S) = 1$  holds identically along the boundary, so differentiating and using  $V''(y^*) = 0$  gives  $\partial V' / \partial \lambda_S(y^*) = 0$ , that is

$$W'(y^*) = 0. \quad (19)$$

At the recapitalisation trigger, differentiating the value-matching condition  $V(x_L) = \mathcal{M}V(x_L)$  along the boundary, with the first-order terms cancelling by the impulse smooth fit  $V'(x_L) = (\mathcal{M}V)'(x_L) = 1 + \kappa$  and the  $y_{\text{post}}$  term vanishing by the envelope property of the inner supremum, gives the nonlocal condition

$$W(x_L) = W(y_{\text{post}}). \quad (20)$$

The second-order smooth fit  $V''(y^*) = 0$  is what makes the barrier tractable rather than what obstructs it: it is precisely the condition delivering the Neumann datum (19).

**Lemma 3** (BDR: Differentiability of the boundaries). *Let  $\theta$  be either parameter. Suppose  $V$  is  $C^3$  in  $x$  in a neighbourhood of  $y^*$  and  $C^2$  in  $x$  near  $x_L$  and  $y_{\text{post}}$ , with those  $x$ -derivatives jointly continuous in  $(x, \theta)$  and continuously differentiable in  $\theta$ . Then, on the set where the corresponding non-degeneracy condition holds,*

- (i)  $y^*(\theta)$  is continuously differentiable wherever  $\rho_L \neq \mu'(y^*)$ ;
- (ii)  $x_L(\theta)$  is continuously differentiable wherever  $V''(x_L) \neq 0$ ;
- (iii)  $y_{\text{post}}(\theta)$  is continuously differentiable wherever  $V''(y_{\text{post}}) \neq 0$ .

*Proof.* Each boundary is the zero set of a scalar function to which the implicit function theorem applies; in each case the required non-degeneracy is the  $x$ -derivative of that function.

For  $y^*$ , the defining relation is not  $V'(y) - 1 = 0$  — that equation is degenerate here, since  $V''(y^*) = 0$  — but the second-order smooth fit itself. Set  $F(y, \theta) := V''(y; \theta)$ , so  $F(y^*(\theta), \theta) = 0$ . Then  $\partial F / \partial y = V'''(y^*)$ , and evaluating the differentiated interior equation at  $y^*$ , where  $V'' = 0$ ,  $V' = 1$  and  $\Lambda' = 0$ , gives  $\frac{1}{2}\sigma^2(y^*)V'''(y^*) = \rho_L - \mu'(y^*)$ . Since  $\sigma^2(y^*) > 0$ ,  $\partial F / \partial y \neq 0$  exactly when  $\rho_L \neq \mu'(y^*)$ , which is (i).

For  $x_L$ , set  $G(x, \theta) := V'(x; \theta) - (1 + \kappa)$ , so  $G(x_L(\theta), \theta) = 0$  by the impulse smooth fit, and  $\partial G / \partial x = V''(x_L)$ , giving (ii). For  $y_{\text{post}}$  the same function serves, with the first-order condition  $V'(y_{\text{post}}) = 1 + \kappa$  in place of the smooth fit and  $\partial G / \partial x = V''(y_{\text{post}})$ , giving (iii).  $\square$

The three non-degeneracy conditions in Lemma 3 are not new hypotheses: they are the same conditions —  $\rho_L > \mu'(y^*)$ ,  $V''(x_L) > 0$ ,  $V''(y_{\text{post}}) < 0$  — that sign the derivatives in Propositions 12 and 13. Each therefore does double duty, delivering both the existence of the derivative and its sign. What remains genuinely assumed is the joint regularity of  $V$  in  $(x, \theta)$  stated at the head of the lemma, which is a regularity question for the quasi-variational inequality itself and is not settled here.

**Proposition 12** (TCS: Threshold comparative statics). *Assume Lemmas 1–3, work at a  $\lambda_S$  of differentiability, and take  $V''(x_L) > 0$  from Proposition 14(ii). Then*

$$\frac{dy^*}{d\lambda_S} = \frac{\rho_L W(y^*)}{\rho_L - \mu'(y^*)}, \quad \frac{dx_L}{d\lambda_S} = \frac{W'(x_L)}{V''(x_L)}. \quad (21)$$

*Consequently:*

- (i)  $dy^*/d\lambda_S > 0$  whenever  $W(y^*) > 0$  and  $\rho_L > \mu'(y^*)$ , the latter being equivalent to  $\rho > r + \pi(\mu_s - r)$  at  $y^*$ ;
- (ii)  $dx_L/d\lambda_S > 0$  whenever  $\rho_L W(y_{\text{post}}) < R - x_L$ .

*Proof. Barrier.* Differentiating  $V'(y^*(\lambda_S); \lambda_S) = 1$  gives  $V''(y^*) dy^*/d\lambda_S + \partial V' / \partial \lambda_S(y^*) = 0$ , which is  $0 = 0$  by  $V''(y^*) = 0$  and (19): the first-order condition is degenerate and carries no information. Differentiate instead the second-order condition  $V''(y^*(\lambda_S); \lambda_S) = 0$ , giving  $V'''(y^*) dy^*/d\lambda_S = W''(y^*)$ , since  $\partial V'' / \partial \lambda_S = -W''$ . Evaluate the two third-order quantities. In (18) at  $y^*$ , using  $W'(y^*) = 0$  and  $(R - y^*)^+ = 0$  for  $y^* > R$ , one gets  $\frac{1}{2}\sigma^2(y^*)W''(y^*) = \rho_L W(y^*)$ . Differentiating the interior equation and evaluating at  $y^*$ , where  $V'' = 0$ ,  $V' = 1$  and  $\Lambda' = 0$ , gives  $\frac{1}{2}\sigma^2(y^*)V'''(y^*) = \rho_L - \mu'(y^*)$ . Taking the ratio, the common factor  $\frac{1}{2}\sigma^2(y^*)$  cancels and the first identity in (21) follows. For (i),  $W \geq 0$  always and  $W(y^*) > 0$  whenever the buffer reaches deficit with positive probability; and since  $\mu'(y^*) = \mu_{\text{opt}}(y^*) - \mu_L$  with  $\mu_{\text{opt}} = r + \pi(\mu_s - r)$ , the condition  $\rho_L > \mu'(y^*)$  is  $\rho > \mu_{\text{opt}}(y^*)$ .

*Trigger.* Differentiating the impulse smooth fit  $V'(x_L(\lambda_S); \lambda_S) = 1 + \kappa$  gives  $V''(x_L) dx_L/d\lambda_S + \partial V' / \partial \lambda_S(x_L) = 0$ , and  $\partial V' / \partial \lambda_S = -W'$ , yielding the second identity in (21). For the sign, let  $\tau$  be the first hitting time of  $x_L$  from  $x > x_L$  and decompose (16) at  $\tau$ , using that the policy injects at  $x_L$  and restarts from  $y_{\text{post}}$ :

$$W(x) - W(y_{\text{post}}) = \mathbb{E}_x \left[ \int_0^\tau e^{-\rho_L t} (R - X_t)^+ dt \right] - (1 - \mathbb{E}_x[e^{-\rho_L \tau}]) W(y_{\text{post}}).$$

As  $x \downarrow x_L$  we have  $\tau \rightarrow 0$ , and to first order in  $\mathbb{E}_x[\tau]$  the right-hand side is  $\mathbb{E}_x[\tau] [(R - x_L) - \rho_L W(y_{\text{post}})] + o(\mathbb{E}_x[\tau])$ , using  $\mathbb{E}_x[\int_0^\tau e^{-\rho_L t} (R - X_t)^+ dt] = (R - x_L)\mathbb{E}_x[\tau] + o(\mathbb{E}_x[\tau])$  and  $1 - \mathbb{E}_x[e^{-\rho_L \tau}] = \rho_L \mathbb{E}_x[\tau] + o(\mathbb{E}_x[\tau])$ . Since  $\mathbb{E}_x[\tau]$  is strictly increasing in  $x$  near  $x_L$  and vanishes at  $x_L$ , the sign of  $W'(x_L)$  is that of  $(R - x_L) - \rho_L W(y_{\text{post}})$ . With  $V''(x_L) > 0$  this gives (ii).  $\square$

*Remark 1* (TCSN: What the proposition rests on). The differentiability of  $V$  in  $\lambda_S$  and of the two boundaries is established, not assumed: the first by Lemma 1 (the value is convex in each parameter, hence differentiable off a countable set), the second by Lemma 3 (the implicit function theorem, whose non-degeneracy conditions are the sign conditions below). The hypothesis  $V''(x_L) > 0$  is Proposition 14(ii), which supplies it from the variational inequality up to the non-degeneracy  $\Psi(x_L) > 0$ ; it is no longer the numerical finding of Remark 9. What the solve adds is not confirmation of the identities —  $W$  is measured there as  $-\partial V/\partial \lambda_S$ , so (17) is definitional in the numerics and the barrier identity is a rearrangement of the barrier condition itself — but a check of the two sign conditions. At the stated parameters  $\rho_L - \mu'(y^*) = 0.045 > 0$ , and condition (i) is implied by the parameter-validity requirement  $\rho > r + \mu_r^+/\bar{a}$  already imposed on the solve, since the cap bounds  $\pi(\mu_s - r) \leq \mu_r^+/\bar{a}$ . Condition (ii) holds across the sweep with  $R - x_L$  exceeding  $\rho_L W(y_{\text{post}})$  by a factor between 11 and 28, the margin widening in  $\lambda_S$ . Both are therefore slack rather than marginal at the calibration, though neither is guaranteed in general: (ii) fails when recapitalisation barely reduces continuation shortfall, and in that regime the trigger need not rise.

#### 4.1 The fixed cost, and joint identification

The same construction applies to the fixed issuance cost. Like  $\lambda_S$ , the parameter  $K$  enters the objective linearly and never enters the admissible set, so the argument of Lemma 2 transfers verbatim with one substitution: the running shortfall is replaced by the discounted *count* of injections,

$$N(x) := \mathbb{E}_x \left[ \sum_i e^{-\rho_L \tau_i} \right], \quad (22)$$

the sum over injection times  $\tau_i$  under the optimal policy.

**Lemma 4** (ENVK: The envelope identity for the fixed cost). *At any  $K$  at which  $V(x; \cdot)$  is differentiable — by Lemma 1(ii), all but countably many —*

$$\frac{\partial V}{\partial K}(x) = -N(x), \quad (23)$$

*and on the inaction region  $N$  satisfies the homogeneous equation*

$$\frac{1}{2}\sigma^2(x, u(x))N''(x) + \mu(x, u(x))N'(x) - \rho_L N(x) = 0, \quad (24)$$

*with boundary data  $N'(y^*) = 0$  and  $N(x_L) = N(y_{\text{post}}) + 1$ .*

*Proof.* Write  $J(x, a; K) = B(x, a) - K N(x, a)$ , where  $N(x, a)$  is the discounted injection count under  $a$  and  $B$  collects the terms not involving  $K$ . The decomposition is exact because  $K$  multiplies the number of impulses and nothing else, and it does not restrict the admissible set; Danskin's theorem then gives (23) as in Lemma 2. Equation (24) is homogeneous rather than inhomogeneous because injections accrue only at boundary hits: no injection occurs while the state is interior, so there is no running term. The Neumann datum at  $y^*$  follows from differentiating  $V'(y^*(K); K) = 1$  and using  $V''(y^*) = 0$ , exactly as (19) did. At the trigger, differentiate  $V(x_L) = V(y_{\text{post}}) - K - (1 + \kappa)(y_{\text{post}} - x_L)$  along the boundary; the terms in  $y'_{\text{post}}$  cancel by the first-order condition  $V'(y_{\text{post}}) = 1 + \kappa$ , the terms in  $x'_L$  cancel by the smooth fit  $V'(x_L) = 1 + \kappa$ , and the explicit  $-\partial K/\partial K = -1$  survives, giving  $\partial V/\partial K(x_L) = \partial V/\partial K(y_{\text{post}}) - 1$ , which is the stated condition. It also follows directly from (22): from  $x_L$  the policy injects at once and then faces the continuation count from  $y_{\text{post}}$ .  $\square$

**Proposition 13** (KCS: Fixed-cost comparative statics). *Assume Lemma 4 and Lemmas 1–3, work at a  $K$  of differentiability, and take  $V''(x_L) > 0$  and  $V''(y_{\text{post}}) < 0$  from Proposition 14. Then*

$$\frac{dy^*}{dK} = \frac{\rho_L N(y^*)}{\rho_L - \mu'(y^*)} > 0, \quad \frac{dx_L}{dK} = \frac{N'(x_L)}{V''(x_L)} < 0, \quad \frac{dy_{\text{post}}}{dK} = \frac{N'(y_{\text{post}})}{V''(y_{\text{post}})} > 0, \quad (25)$$

*the first under  $\rho_L > \mu'(y^*)$ . In particular the inaction region widens at both ends, and the trigger-to-target gap  $y_{\text{post}} - x_L$  is strictly increasing in  $K$ .*

*Proof.* For  $x$  above the trigger, conditioning on the first hitting time  $\tau$  of  $x_L$  gives  $N(x) = \mathbb{E}_x[e^{-\rho_L \tau}](1 + N(y_{\text{post}}))$ , and  $x \mapsto \mathbb{E}_x[e^{-\rho_L \tau}]$  is strictly decreasing, so  $N' < 0$  throughout the inaction region and  $N > 0$ . The barrier identity is obtained exactly as in Proposition 12: the first-order condition is degenerate, and differentiating  $V''(y^*(K); K) = 0$  gives  $V'''(y^*) dy^*/dK = N''(y^*)$ ; evaluating (24) at  $y^*$  with  $N'(y^*) = 0$  gives  $\frac{1}{2}\sigma^2 N''(y^*) = \rho_L N(y^*)$ , and  $\frac{1}{2}\sigma^2 V'''(y^*) = \rho_L - \mu'(y^*)$  as before. The trigger identity follows from differentiating the smooth fit  $V'(x_L(K); K) = 1 + \kappa$ , and is negative since  $N'(x_L) < 0$  and  $V''(x_L) > 0$ . The target identity follows from differentiating the first-order condition  $V'(y_{\text{post}}(K); K) = 1 + \kappa$ , and is positive since  $N'(y_{\text{post}}) < 0$  and  $V''(y_{\text{post}}) < 0$  in the concave interior. The gap statement is the difference of the last two.  $\square$

**Corollary 2** (IDN: Joint local identification). *Under Propositions 12 and 13 and their sign conditions, the Jacobian of  $(y^*, x_L)$  with respect to  $(\lambda_S, K)$  is nonsingular:*

$$\det \frac{\partial(y^*, x_L)}{\partial(\lambda_S, K)} = \frac{\rho_L}{(\rho_L - \mu'(y^*))V''(x_L)} \left[ W(y^*)N'(x_L) - N(y^*)W'(x_L) \right] < 0. \quad (26)$$

*Hence  $(\lambda_S, K)$  is locally identified from the pair of thresholds.*

*Proof.* The matrix entries are those of (21) and (25); the common factors  $\rho_L/(\rho_L - \mu'(y^*))$  and  $1/V''(x_L)$  factor out of the first row and second row respectively. In the bracket,  $W(y^*) > 0$  and  $N'(x_L) < 0$  make the first product negative, while  $N(y^*) > 0$  and  $W'(x_L) > 0$  make the second positive, so the bracket is strictly negative; the prefactor is positive. The determinant is therefore strictly negative, in particular nonzero, and the inverse function theorem applies.  $\square$

**Remark 2** (KCSN: Reading the identification result). The two parameters separate for a structural reason rather than a numerical one:  $\lambda_S$  and  $K$  both raise the payout barrier, but they move the recapitalisation trigger in *opposite* directions — a steeper shortfall penalty makes the firm intervene earlier, a larger fixed cost makes it wait longer. That opposition is what signs (26) away from zero, and it is why the determinant does not depend on comparing magnitudes. The claim is local identification from the thresholds, not a statement that any estimator attains it.

Two cautions, and one piece of support. The result inherits every hypothesis of the two propositions, including  $V''(x_L) > 0$ , now Proposition 14(ii) rather than a numerical finding; and local identification from the thresholds is not a claim that any particular estimator attains it. The support is that the converged  $K$ -sweep of Remark 10, run at  $\lambda_S = 1$  across  $K \in \{0, \dots, 0.05\}$  with every run converging, already exhibits both predicted directions. The continuation band  $y^* - x_L$  widens monotonically from 0.159 to 0.685, which is  $dy^*/dK - dx_L/dK > 0$ ; and at  $K = 0.05$  the recapitalisation region is one cell wide with the trigger collapsed onto the lower boundary, which is  $dx_L/dK < 0$  directly. Combining the two — a band of 0.685 with the trigger at the boundary places  $y^*$  near 1.69, against 1.5715 at  $K = 0.02$  — indicates  $dy^*/dK > 0$  as well, though that last step chains two separately reported quantities rather than reading a recorded  $y^*$ , and a sweep recording both thresholds directly would settle it without the inference.

## 5 Second-order conditions at the impulse boundaries

The hypotheses  $V''(x_L) > 0$  and  $V''(y_{\text{post}}) < 0$  have so far been carried as numerical findings. Both are in fact second-order conditions, and both follow from the structure of the impulse operator once one observation is made:  $\mathcal{M}v$  is *affine*. Writing the injection in terms of the target  $y = x + \xi$ ,

$$\mathcal{M}v(x) = \sup_{y>x} \{V(y) - (1 + \kappa)(y - x)\} - K = (1 + \kappa)x - K + \sup_y \{V(y) - (1 + \kappa)y\}, \quad (27)$$

the supremum on the right being independent of  $x$ . Two consequences follow at once. The injection target does not depend on where the injection is made from, so  $y_{\text{post}}$  is a single number rather than a function of  $x$ ; and  $g := \mathcal{M}v - V$  satisfies  $g' = (1 + \kappa) - V'$  and  $g'' = -V''$ .

Write  $\varphi(y) := V(y) - (1 + \kappa)y$ , so that  $y_{\text{post}}$  maximises  $\varphi$  and  $\varphi' = V' - (1 + \kappa)$ . It is convenient to record the sign test the interior equation supplies. At any point where  $V' = 1 + \kappa$ , substituting into  $\frac{1}{2}\sigma^2 V'' + \mu V' - \rho_L V - \Lambda = 0$  gives

$$\frac{1}{2}\sigma^2(x, u(x)) V''(x) = \Psi(x), \quad \Psi(x) := \rho_L V(x) + \Lambda(x) - (1 + \kappa)\mu(x, u(x)), \quad (28)$$

so at such a point  $V''$  and  $\Psi$  carry the same sign.

**Proposition 14** (SOC: Second-order conditions). *Assume  $V' > 1 + \kappa$  on  $(x_L, y_{\text{post}})$  and  $V' < 1 + \kappa$  on  $(y_{\text{post}}, y^*)$ , and, for the second half of (ii) only, the smooth fit  $V'(x_L) = 1 + \kappa$  — which is Proposition 15, proved independently below. Then*

- (i)  $V''(y_{\text{post}}) < 0$ , strictly;
- (ii)  $V''(x_L^+) \geq 0$ , with equality if and only if the interior equation holds with equality at  $x_L$  from the intervention side; equivalently,  $V''(x_L^+) > 0$  if and only if  $\Psi(x_L) > 0$ .

*Proof.* (i) By hypothesis  $\varphi' > 0$  on  $(x_L, y_{\text{post}})$  and  $\varphi' < 0$  on  $(y_{\text{post}}, y^*)$ , so  $\varphi'$  changes sign at  $y_{\text{post}}$  and its zero there has odd order. If  $\varphi''(y_{\text{post}}) = 0$  then,  $\varphi$  being  $C^3$  near  $y_{\text{post}}$  by Lemma 5,  $\varphi'$  would vanish to order at least two, and a zero of even order admits no sign change — a contradiction unless  $\varphi'$  vanishes identically nearby, which the hypothesis excludes. Hence  $\varphi''(y_{\text{post}}) \neq 0$ ; being a maximum it satisfies  $\varphi'' \leq 0$ , so  $V''(y_{\text{post}}) = \varphi''(y_{\text{post}}) < 0$ .

(ii) On the intervention region  $V = \mathcal{M}V$ , which by (27) is affine with slope  $1 + \kappa$ ; hence  $V' = 1 + \kappa$  and  $V'' = 0$  there. The QVI requires the first branch of (8) to be non-positive wherever the impulse branch is active, so for  $x \leq x_L$

$$\frac{1}{2}\sigma^2 \cdot 0 + \mu(x)(1 + \kappa) - \rho_L V(x) - \Lambda(x) \leq 0,$$

which is exactly  $\Psi(x) \geq 0$  on the intervention region, and by continuity  $\Psi(x_L) \geq 0$ . This much uses only the variational inequality on the intervention side; it is what Proposition 15 consumes, and it is why that proposition does not depend on the remainder of this one. Since  $V'(x_L) = 1 + \kappa$  by Proposition 15, (28) applies at  $x_L$  and gives  $\frac{1}{2}\sigma^2 V''(x_L^+) = \Psi(x_L) \geq 0$ , so  $V''(x_L^+) \geq 0$  with equality precisely when  $\Psi(x_L) = 0$ , that is when the displayed inequality binds. The same conclusion follows without (28) from  $g \leq 0$ ,  $g(x_L) = 0$  and  $g'(x_L) = 0$ , which force  $g''(x_L^+) \leq 0$ .  $\square$

*Remark 3* (SOCN: Strictness at the trigger). Part (i) is unconditional given the hypotheses: nothing numerical enters. Part (ii) delivers the weak inequality unconditionally and locates strictness in a single interpretable condition — that waiting is *strictly* suboptimal just below the trigger, rather than marginally so. That is the natural non-degeneracy for an intervention region with non-empty interior, and it fails only in the knife-edge case where the firm is indifferent between acting

and waiting on a whole neighbourhood. At the numerical baseline  $\Psi$  is bounded away from zero throughout the intervention region, with  $\Psi(x_L)$  equal to 0.012, 0.023 and 0.035 at  $\lambda_S = 1, 1.25$  and 2 respectively, rising in  $\lambda_S$ ; the corresponding  $V''(x_L^+)$  are 6.4, 13.9 and 18.7, of the same sign in every case as (28) requires.

The consequence for the results above is that Remark 9 is no longer load-bearing. Propositions 12 and 13, Corollary 2, Lemma 3 and Proposition 16 took  $V''(x_L) > 0$  and  $V''(y_{\text{post}}) < 0$  as hypotheses supported by a numerical finding; they now rest on Proposition 14, which supplies the first up to a non-degeneracy condition and the second outright. What those results still inherit is the smooth fit at  $x_L$  — no longer an assumption either, but Proposition 15 below.

**Proposition 15** (SMF: Smooth fit at the trigger). *Let  $V$  be the value function and suppose  $\sigma^2(x_L, \pi) > 0$  for the optimal  $\pi$  at  $x_L$ , which holds since  $x_L > 1$  (Lemma 5). Then  $V$  is differentiable at  $x_L$ , with*

$$V'(x_L^-) = V'(x_L^+) = 1 + \kappa.$$

*Proof.* Write  $J := V'(x_L^+) - V'(x_L^-)$ , both one-sided derivatives existing by convexity of  $V - \mathcal{M}V$  near  $x_L$  and interior regularity above it.

*Step 1:*  $J \geq 0$ , and the left derivative is exactly  $1 + \kappa$ . On the intervention region  $V = \mathcal{M}V$ , which by (27) is affine with slope  $1 + \kappa$ ; hence  $V'(x_L^-) = 1 + \kappa$ . The obstacle  $g := V - \mathcal{M}V$  satisfies  $g \geq 0$  with  $g(x_L) = 0$ , so  $x_L$  minimises  $g$  from the right and  $g'(x_L^+) \geq 0$ , that is  $V'(x_L^+) \geq 1 + \kappa$ . Any kink at  $x_L$  is therefore convex. Note this uses only the variational inequality, not smooth fit, so no circularity arises.

*Step 2:*  $J > 0$  contradicts the supersolution property.  $V$  is a viscosity solution of (8); in particular a supersolution, so for every  $\varphi \in C^2$  touching  $V$  from below at  $x_L$ ,

$$\min \left\{ -(\mathcal{L}^\pi \varphi - \rho_L V - \Lambda), \varphi' - 1, V - \mathcal{M}V \right\}(x_L) \geq 0.$$

Since  $V(x_L) = \mathcal{M}V(x_L)$  the third entry vanishes, so the first must be non-negative. Suppose  $J > 0$  and write  $\varphi(x) = V(x_L) + p(x - x_L) + \frac{1}{2}q(x - x_L)^2 + o((x - x_L)^2)$ . For  $\varphi \leq V$  near  $x_L$  it suffices that  $p \in (1 + \kappa, 1 + \kappa + J)$ : on each side  $V - \varphi$  then has a strictly signed linear term, which dominates the quadratic for small  $|x - x_L|$  whatever the value of  $q$ . Thus  $q$  may be taken arbitrarily large. But

$$-(\mathcal{L}^\pi \varphi - \rho_L V - \Lambda)(x_L) = -\frac{1}{2}\sigma^2(x_L, \pi)q - \mu(x_L, \pi)p + \rho_L V(x_L) + \Lambda(x_L),$$

which tends to  $-\infty$  as  $q \rightarrow \infty$  because  $\sigma^2(x_L, \pi) > 0$ . The supersolution inequality fails, so  $J > 0$  is impossible.

With Step 1 this forces  $J = 0$ . □

*Remark 4* (SMFN: Why the argument does not prove too much). The same computation run at  $J = 0$  is consistent, which is the check that the argument discriminates rather than merely destroys. If  $V$  is differentiable at  $x_L$  then any  $\varphi$  touching from below must satisfy  $\varphi'(x_L) = 1 + \kappa$ , and since  $V$  is affine on the intervention side,  $\varphi \leq V$  with matching value and slope forces  $\varphi''(x_L) \leq 0$ . The first entry is then bounded below by  $\rho_L V + \Lambda - (1 + \kappa)\mu = \Psi(x_L)$ , non-negative by the intervention-region inequality established in the proof of Proposition 14(ii) — a step that itself uses only the variational inequality, so nothing here is circular. The supersolution property holds. What breaks the kinked case is not curvature but the freedom a corner creates: when  $V'$  jumps, a test function may match neither one-sided slope, and that decouples  $\varphi''$  from any constraint.

Proposition 15 supplies hypothesis (H1) of Proposition 16 at the one point where it was previously asserted. Interior regularity gives the rest of (H1) (Lemma 5), and (H2)–(H4) were already established, so verification is now unconditional.

## 6 Regularity and verification

The comparative statics of Section 4 differentiate  $V$ , and Lemma 3 assumes joint regularity. This section records what that regularity amounts to, and what a verification theorem would and would not require. It closes neither question, but it locates both more precisely than an assumption does.

**Lemma 5** (ITR: Interior regularity). *On any compact subset of  $(1, \infty)$  the diffusion is uniformly elliptic:  $\sigma^2(x, \pi) \geq (1 - |c|)[(\pi\sigma x)^2 + \sigma_L^2(1 - x)^2] > 0$  for  $|c| < 1$  and  $x > 1$ . On the inaction region the equation is the linear second-order ODE  $\frac{1}{2}\sigma^2(x, u(x))v'' + \mu(x, u(x))v' - \rho_L v = \Lambda(x)$ , whose coefficients are Lipschitz and are smooth away from the finitely many points where the cap switches regime or  $\Lambda$  kinks. Hence  $v \in C^2$  on the inaction region and  $v \in C^3$  away from  $\{R, \bar{x}\}$ , where  $\bar{x}$  is the solvency-liquidity crossing.*

*Proof.* The quadratic form  $a^2 + 2cab + b^2$  with  $a = \pi\sigma x$  and  $b = \sigma_L(1 - x)$  is bounded below by  $(1 - |c|)(a^2 + b^2)$ , which is positive for  $x > 1$  since then  $b \neq 0$ . Uniform ellipticity on compacts plus Lipschitz coefficients gives  $v \in C_{\text{loc}}^{2,\alpha}$  by standard linear theory; where the coefficients are additionally differentiable, differentiating the equation gives  $v'''$  in terms of lower derivatives, so  $v \in C^3$  there. The coefficients fail to be differentiable only where  $u(\cdot)$  switches between the two branches of the cap, at  $\bar{x}$ , and where  $\Lambda$  kinks, at  $R$ .  $\square$

Two consequences matter for Lemma 3. The ellipticity degenerates only at  $x = 1$ , and only because the cap drives  $\pi$  to zero there; the state never reaches that boundary (Corollary 1), so the degeneracy is not encountered under any admissible policy. And the  $C^3$  requirement near  $y^*$  is met whenever  $y^*$  is separated from  $\{R, \bar{x}\}$ , which at the numerical baseline it is:  $R = 1.15$  and  $\bar{x} \approx 1.167$ , against  $y^* \approx 1.57$ . The  $C^3$  requirement is one-sided — it applies to the inaction-region solution extended to  $y^*$ , not across it, since  $v$  is linear above  $y^*$  and  $v'''$  jumps there by construction.

**Lemma 6** (FIN: No chattering). *For  $K > 0$ , any admissible strategy with finite value makes only finitely many impulses in expectation, discounted: the discounted count  $N$  of (22) is finite. Consequently the sum over impulses in the verification argument converges absolutely.*

*Proof.* Each impulse costs at least  $K > 0$ . A strategy making impulses at times  $\tau_i$  incurs discounted cost at least  $K \sum_i e^{-\rho_L \tau_i} = KN$ . The value of any strategy is bounded above by the value of the unconstrained problem, which is finite under the growth condition below, so  $KN < \infty$  and hence  $N < \infty$ . The fixed cost, which obstructs concavity, is what rules out chattering.  $\square$

**Proposition 16** (VER: Verification). *Let  $v$  satisfy: (H1)  $v \in C^1$  on  $(1, \infty)$  and  $C^2$  off a finite set; (H2) the three QVI inequalities of (8) pointwise; (H3) linear growth,  $|v(x)| \leq C(1 + x)$ ; and (H4)  $\rho > \sup_x \mu_{\text{opt}}(x)$ . Then  $v \geq J(\cdot, a)$  for every admissible  $a$ , with equality for the barrier-trigger policy; hence  $v = V$  and that policy is optimal.*

*Proof sketch.* Apply Itô to  $e^{-\rho_L t} v(X_t)$  under an arbitrary admissible  $a$ , splitting the drift, the singular and the impulse parts. The drift term is non-positive by the first inequality in (H2), since it holds for every  $\pi \leq \bar{\pi}(x)$  and in particular for the one used. The singular term contributes  $\int e^{-\rho_L t} v'(X_t) dD_t \geq \int e^{-\rho_L t} dD_t$  by  $v' \geq 1$ . Each impulse contributes  $v(x + \xi) - v(x) \leq K + (1 + \kappa)\xi$  by  $v \geq \mathcal{M}v$ , and the sum over impulses converges by Lemma 6. Taking expectations and letting  $T \rightarrow \infty$ , the boundary term  $\mathbb{E}[e^{-\rho_L T} v(X_T)]$  vanishes by (H3) and (H4): the drift is bounded above by  $x(\mu_{\text{opt}} - \mu_L) + \gamma$ , so  $X$  grows at most at rate  $\sup \mu_{\text{opt}} - \mu_L$ , and (H4) is exactly  $\rho_L > \sup \mu_{\text{opt}} - \mu_L$ . Reversing the inequalities for the candidate policy, which attains equality in each, gives the result.  $\square$

*Remark 5* (VERN: What verification needs). Concavity of  $V$  is not among the hypotheses, and its absence (Remark 9) is therefore not an obstruction to verification. Concavity is normally used to *derive* the threshold form of the optimal policy a priori; here that form is imposed by construction and validated ex post, so the role concavity would play is already discharged. What is needed in its place is local:  $V''(y_{\text{post}}) < 0$ , so the maximiser in  $\mathcal{M}v$  is unique — a strictly weaker requirement than global concavity, and the same hypothesis Proposition 13 already carries. The fixed cost, meanwhile, cuts both ways: it destroys concavity but supplies Lemma 6, without which the impulse sum need not converge.

Hypothesis (H4) is the same condition met twice already — it signs  $dy^*/d\lambda_S$  in Proposition 12, supplies the non-degeneracy for Lemma 3(i), and here delivers transversality. It is enforced on every solve by the parameter-validity check of Remark 6.

Hypothesis (H1) is now established rather than assumed. Interior regularity (Lemma 5) gives  $C^2$  and locally  $C^3$  away from the free boundaries; the  $C^2$  fit at  $y^*$  is the smooth-fit condition itself; value matching at  $x_L$  gives  $C^0$ ; and the  $C^1$  fit at  $x_L$ , the one point previously asserted, is Proposition 15. Proposition 16 is therefore unconditional.

It is worth recording why the numerics could not settle this and the argument could. The discrete scheme enforces  $v = \max(v, \mathcal{M}v)$  pointwise, so the computed solution carries a corner at whichever node the boundary falls on, at every mesh; the observed one-sided slope gap is consistent with a genuine kink and with a steep continuous rise alike, and separating them requires resolving a layer that the affordable meshes do not resolve. The supersolution argument avoids the question entirely: it does not measure the corner, it shows a corner cannot exist.

## 7 The numerical solve

This section records what a numerical solve of (8) establishes. Its contents are findings at stated parameters, not theorems. All figures use  $r = 0.025$ ,  $\mu_s = 0.04$ ,  $\mu_L = 0.03$ ,  $\gamma = 0.02$ ,  $\rho = 0.12$  (so  $\rho_L = 0.09$ ),  $\sigma = 0.08$ ,  $\sigma_L = 0.03$ ,  $c = 0.20$ ,  $a_1 = 0.045$ ,  $a_2 = 0.05$ ,  $a_3 = 0.30$ ,  $K = 0.02$ ,  $\kappa = 0.01$ ,  $R = 1.15$ , on a uniform grid with  $y_{\text{max}} = 2.5$  and  $N = 301$  unless stated otherwise.

*Remark 6* (SLV: The solver, and how it was validated). The vehicle is the benchmark’s own published implementation. Three validations were run before any conclusion was drawn from it. First, *reproduction*: at the benchmark’s own baseline it returns  $y^* = 1.673866$ ,  $y_{\text{post}}^* = 1.227682$  and a value uplift of 0.522875, against published 1.674, 1.228 and 0.5229 — four significant figures on all three. Second, *nesting as a software test*:  $\Lambda$  enters the equation linearly, so it was added as a source term on the implicit step, and Proposition 6 then supplies a falsifiable check — at  $\lambda_S = 1$  the modified solver must reproduce the unmodified one *exactly*. It does, to a difference of zero, and at  $\lambda_S = 1.5$  the value function is strictly lower below  $R$ , confirming the term is active rather than silently dropped. Third, *operator verification*: the impulse operator  $\mathcal{M}$  is implemented as an  $O(N)$  suffix maximum, and is checked against a direct  $O(N^2)$  supremum over the same grid on six profiles — concave, affine, non-concave in two distinct shapes, piecewise-flat, and monotone decreasing — agreeing to  $5.6 \times 10^{-16}$ . The non-concave profiles are the point of the check: a suffix maximum is most likely to misbehave exactly where  $V$  is not concave, which is the regime Remark 9 reports. Small grids, degenerate cost parameters, the strictness of  $\xi > 0$  and the sentinel at  $y_{\text{max}}$  are covered by separate tags. Convergence was certified by re-solving every  $\lambda_S$  at a doubled iteration cap. All six then converge, at 1645–1669 iterations, and the induced shifts in  $y^*$  fall monotonically from  $5.0 \times 10^{-4}$  to  $4.1 \times 10^{-4}$  across the range — 4.9% of the smallest inter-row gap, so the orderings below survive with about a twentyfold margin. The recapitalisation boundary is unmoved to machine precision, being pinned to the mesh. All figures quoted here are from the converged solves.

*Remark 7* (CBI: The cap binds throughout the inaction region). Under the solved policy the regulatory constraint is active across the whole inaction region  $[x_L, y^*]$ : comparing the solver’s chosen  $\pi^*$  against  $\bar{\pi}$  cell by cell, the two agree at every one of the 108–111 cells at  $\lambda_S \in \{1.00, 1.10, 1.25, 1.50, 1.75\}$ . At  $\lambda_S = 2.00$  three cells show a slack cap, and they straddle the regime switch  $\bar{x} = 1.167647$ , so they are more plausibly an artefact of the switch than an interior slack region. The comparison is a genuine test rather than a tautology: the solver searches  $\pi$  over a grid spanning  $[0, \bar{\pi}(x)]$  and retains the maximiser, so a slack optimum would be found if one existed.

This is a finding at these parameters and not a theorem, and it should be read at that strength. What it establishes is that the capped coefficients (11) describe the whole of the inaction region here, and that what closes the problem is the impulse control rather than any interior relaxation of the cap. The formulation (8) accommodates either outcome without modification.

*Remark 8* (CSL: Comparative statics in  $\lambda_S$ ). The recapitalisation trigger is monotone in  $\lambda_S$ , and so is the dividend barrier. Under the benchmark’s proportional-cost operator  $\mathcal{H}$ , as  $\lambda_S$  runs over  $\{1, 1.1, 1.25, 1.5, 1.75, 2\}$ :  $y^*$  rises  $1.3725 \rightarrow 1.4371$ , the recapitalisation boundary rises  $1.145 \rightarrow 1.200$ , and  $V(1)$  falls  $0.5037 \rightarrow 0.4702$ . Under the additive-cost operator  $\mathcal{M}$  the same orderings hold and are larger:  $y^*$  rises  $1.5715 \rightarrow 1.6596$  (a move of 0.088 against 0.065), the boundary rises  $1.030 \rightarrow 1.105$ , and  $V(1)$  falls  $0.3843 \rightarrow 0.3395$ . Every sequence is strictly monotone with no reversal. The economic reading is that a more asymmetric institution retains a larger buffer before paying out *and* recapitalises from a higher level of capital, i.e. intervenes earlier; and because the orderings are common to both cost conventions, they are not an artefact of either.

*Remark 9* (LNC: Local non-concavity under a fixed issuance cost). Under  $\mathcal{M}$  the solve exhibits  $V'' > 0$  on an interval immediately above the recapitalisation boundary. Three features of it are established numerically.

*It is not caused by  $\Lambda$ .* It is present at  $\lambda_S = 1$ , where  $\Lambda \equiv 0$  identically and the problem is the risk-neutral benchmark carrying the additive-cost operator (Proposition 6).

*It is not a discretisation artefact.* A kink artefact is pinned to a fixed number of grid points, so its width in  $x$  shrinks with the mesh; a genuine region has a fixed width, so its point count grows. Over three meshes at  $\lambda_S = 1$  the point count went  $6 \rightarrow 12 \rightarrow 24$  — exactly the mesh ratio at both refinements — while the span converged upward (0.0250, 0.0275, 0.0288). The product of count and spacing is 0.0300 at all three meshes, and Richardson extrapolation on the span gives the same limit, so the region has a converged width of 0.0300 and the  $N = 301$  figure understates it by about a sixth.

*Its mechanism is the fixed cost.* The trigger-to-target distance is near-constant at 0.365–0.370 across the sweep, and  $V'$  must return to  $1 + \kappa$  at the target; a fixed cost separates trigger from target and thereby forces an excursion of  $V'$  between them. Local non-concavity from fixed transaction costs is a familiar feature of impulse-control value functions, not a pathology peculiar to this model.

The region is present at every  $\lambda_S$  tested, occupying 5 to 8 grid points at  $N = 301$  — a width of 0.025 to 0.040, with  $V'$  peaking there at 0.121 to 0.133 above  $1 + \kappa$ . So  $V$  is not concave for  $K > 0$  at these parameters. Mesh refinement across  $\lambda_S$  has been carried out at  $\lambda_S = 1$  only.

*Remark 10* (KSW: The fixed cost is the cause, a  $K$ -sweep). Remark 9 attributes the convex region to the fixed issuance cost  $K$  rather than to  $\Lambda$ . That attribution makes a sharp prediction — remove  $K$  and the region must disappear — and the prediction is testable directly. Fixing  $\lambda_S = 1$ , so that  $\Lambda \equiv 0$  and  $K$  is the only quantity varying, and solving for  $K \in \{0, 0.001, 0.005, 0.01, 0.02, 0.05\}$  at  $N = 301$ , every run converging:

*At  $K = 0$  everything collapses, exactly.* The trigger and the target coincide,  $V'$  never exceeds  $1 + \kappa$  anywhere (the largest excess is  $2 \times 10^{-14}$ , machine zero), the convex region has zero width,

and the one-sided derivatives on both sides of the trigger read  $1 + \kappa$  to eight decimals. Every feature attributed to the fixed cost vanishes precisely when the fixed cost vanishes.

*The classical scaling appears unbidden.* The continuation band  $y^* - x_L$  widens monotonically from 0.159 to 0.685 across the range, and the trigger-to-target distance regressed on  $K$  in logarithms has slope 0.3845, against the  $1/3$  that impulse control with a fixed cost predicts; the ratio of that distance to  $K^{1/3}$  sits between 1.26 and 1.35 over the upper four values. Nothing in the construction was fitted to produce this, so it is an independent check that the solver is doing impulse control rather than something that merely resembles it.

*The width of the convex region grows with  $K$  and then flattens.* Refining to  $N = 601$  and extrapolating gives 0.025 at  $K = 0.005$  and 0.030 at both  $K = 0.01$  and  $K = 0.02$ , the last agreeing exactly with the three-mesh value of Remark 9. Widths read off a single  $N = 301$  mesh are not converged — they err in both directions, by about a sixth at  $K = 0.02$  and about two fifths at  $K = 0.005$  — and an apparent non-monotonicity in them does not survive refinement. Single-mesh widths should not be quoted.

At  $K = 0.05$  the recapitalisation region is one cell wide at  $N = 301$ : the trigger has collapsed onto the lower boundary, and the diagnostics there are degenerate rather than informative. That is economically unsurprising — a large enough fixed cost makes recapitalisation not worth doing until ruin is imminent — but it means the largest  $K$  carries no weight in any of the readings above.

## 8 What is not established

One thing genuinely open, and one place where the numerics mislead about something the theory has since settled.

*The relation between  $\lambda_V$  and  $\lambda_S$ .* Proposition 10 makes this a well-posed question by putting both on the same state measured from the same origin, and Proposition 11 settles that the *definitional* evaluation of  $\lambda_V$  cannot answer it, being independent of  $\lambda_S$ . An *estimated*  $\lambda_V$  does vary with  $\lambda_S$ : applying the panel estimator to simulated model paths (quarterly sampling, sixty quarters, four hundred paths per parameter value) gives a median that falls monotonically from about 1.33–1.39 at  $\lambda_S = 1$  to about 1.15–1.17 at  $\lambda_S = 2$ , depending on the issuance-cost convention. The direction is known, but the movement is small relative to the noise it is measured against: roughly 0.18–0.22 of total change across the full admissible range, against a per-path standard deviation of 0.16–0.20. The estimated ratio is therefore not uninformative but weakly identified — distinguishable from noise only when aggregated across many institutions, not from any single one. Any empirical reading that compares an estimate of  $\lambda_V$  against a benchmark value should report both the direction above and this signal-to-noise magnitude, rather than treat a single point estimate as diagnostic on its own.

*Regularity at the recapitalisation trigger, and why the numerics could not have shown it.* That  $V$  is  $C^1$  at the trigger is settled by Proposition 15, but the route is worth recording because the solve is misleading on the point. For  $K > 0$  the one-sided derivatives at the detected trigger differ in every solve, which would appear to say  $V$  is  $C^0$  but not  $C^1$  there. That reading is an artefact of the scheme: the detected trigger is the last mesh point at which the impulse obstacle is active, so the true boundary lies inside the following cell, and a one-sided difference across that cell mixes the two branches in proportion to where the boundary falls; that proportion moves essentially at random as parameters shift the boundary within a cell, and at fixed mesh and fixed  $K$  the mixed quantity varies by a factor of sixteen across  $\lambda_S$ , non-monotonically, which no genuine property of  $V$  would do. Mesh refinement does not escape this: the scheme enforces  $v = \max(v, \mathcal{M}v)$  pointwise, so a corner sits at whichever node the boundary occupies at every resolution. The question is

therefore not one the mesh can answer, and Proposition 15 answers it instead — not by measuring the corner but by showing, from the variational inequality and the supersolution property, that a corner cannot exist.

## A Verification status

Each result has one canonical verifier: a Lean proof where one exists, an exact symbolic check otherwise. Every Lean declaration cited below is compiler-verified and depends on no `sorry` and no axiom beyond Lean’s classical defaults. Four results are marked *Lean (partial)*: the algebraic core of the argument is compiler-verified with the analytic input (an envelope identity, a cost bound) stated as an explicit hypothesis, and the row records which step remains proved only in the text. Symbolic checks terminate in `simplify(...)==0` or a sign condition on symbols, with no floating point and no sampling. Numerical findings are labelled as such; a numeric check can only fail to find a counterexample and is evidence, not proof.

| Result           | Verifier       | Location                                                                                                                                                                                                                                    |
|------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prop. 1–2        | Lean           | RateBased.lean                                                                                                                                                                                                                              |
| Thm. 1           | symbolic       | verify_rate_based.py (S1a–S1b)                                                                                                                                                                                                              |
| Prop. 3, 4, 5    | symbolic       | verify_rate_based.py                                                                                                                                                                                                                        |
| Prop. 6          | Lean           | QVI_Part1.lean,<br>lambda_asymmetry_vanishes_at_one                                                                                                                                                                                         |
| Prop. 7          | Lean           | QVI_Part1.lean, neg_Lambda_concave                                                                                                                                                                                                          |
| Prop. 8          | symbolic       | verify_state_map.py (M10)                                                                                                                                                                                                                   |
| Prop. 9          | proof in text  | admissibility construction                                                                                                                                                                                                                  |
| Prop. 10, Cor. 1 | symbolic       | verify_state_map.py (M1–M9)                                                                                                                                                                                                                 |
| Prop. 11         | symbolic       | verify_saturation_limits.py (L1–L3;<br>L5* expected-fail)                                                                                                                                                                                   |
| Lem. 1           | proof in text  | convexity of the value in each parameter;<br>differentiable off a countable set                                                                                                                                                             |
| Lem. 2           | Lean (partial) | Envelope.lean, wode_of_envelope: the<br>ODE (18) from the envelope identity (17)<br>as hypothesis; the Danskin step proving<br>(17) is in the text                                                                                          |
| Lem. 3           | proof in text  | implicit function theorem at each bound-<br>ary                                                                                                                                                                                             |
| Prop. 12         | symbolic       | verify_comparative_statics.py (H4a,<br>H4b, H4d)                                                                                                                                                                                            |
| Rem. 1           | numerical      | verify_document_figures.py (F5)                                                                                                                                                                                                             |
| Lem. 4           | Lean (partial) | Envelope.lean, node_of_envelope,<br>neumann_at_ystar, trigger_jump: (24)<br>and both boundary data from (23), the<br>first-order condition, smooth fit and value<br>matching as hypotheses; the Danskin<br>step proving (23) is in the text |
| Prop. 13         | symbolic       | verify_comparative_statics.py (H4c,<br>H4e)                                                                                                                                                                                                 |

| Result   | Verifier       | Location                                                                                                                                                                                                                                                                                                                                  |
|----------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cor. 2   | symbolic       | <code>verify_comparative_statics.py</code> (H5a–H5d)                                                                                                                                                                                                                                                                                      |
| Rem. 2   | numerical      | six $K$ values; directions at stated parameters                                                                                                                                                                                                                                                                                           |
| Prop. 14 | symbolic       | <code>verify_comparative_statics.py</code> (H3a)                                                                                                                                                                                                                                                                                          |
| Rem. 3   | numerical      | <code>verify_document_figures.py</code> (F6–F8)                                                                                                                                                                                                                                                                                           |
| Prop. 15 | Lean           | <code>SmoothFit.lean</code> , <code>smooth_fit</code>                                                                                                                                                                                                                                                                                     |
| Rem. 4   | Lean (partial) | <code>Envelope.lean</code> , <code>smf_permits</code> : every curvature $q \leq 0$ satisfies the operator inequality given $\Psi(x_L) \geq 0$ ; that these are exactly the admissible curvatures under smooth fit is argued in the text                                                                                                   |
| Lem. 5   | proof in text  | interior trigger, $x_L > 1$                                                                                                                                                                                                                                                                                                               |
| Lem. 6   | Lean (partial) | <code>ImpulseCount.lean</code> , <code>count_summable</code> , <code>count_le</code> , <code>impulse_sum_summable</code> : summability of the discounted count with $N \leq B/K$ , and absolute convergence of the impulse sum, from the cost bound as hypothesis; that each impulse costs at least $K$ under finite value is in the text |
| Prop. 16 | proof in text  | Itô argument on $e^{-\rho_L t} v(X_t)$ ; hypotheses (H1)–(H4)                                                                                                                                                                                                                                                                             |
| Rem. 5   | numerical      | <code>verify_smooth_fit_shooting.py</code> (S2–S3)                                                                                                                                                                                                                                                                                        |
| Rem. 6   | numerical      | benchmark reproduced to 4 s.f.; $\lambda_S = 1$ nesting exact; $\mathcal{M}$ vs direct supremum, <code>verify_M_operator.m</code> (V1–V10); the $\mathcal{M}$ sweep convergence-certified                                                                                                                                                 |
| Rem. 7   | numerical      | $\pi^*$ vs $\bar{\pi}$ across the inaction region, six $\lambda_S$                                                                                                                                                                                                                                                                        |
| Rem. 8   | numerical      | six $\lambda_S$ , both operators, strict monotonicity, no reversal; the $\mathcal{H}$ figures are from a sweep not convergence-certified                                                                                                                                                                                                  |
| Rem. 9   | numerical      | mesh-refinement discriminant over three meshes, width converged to 0.0300; present at all six $\lambda_S$                                                                                                                                                                                                                                 |
| Rem. 10  | numerical      | six $K$ values, all converged; $K = 0$ control exact to machine precision; $K^{1/3}$ scaling recovered independently                                                                                                                                                                                                                      |